

## Section 0 – Title and contact information

Title of the proposed solution track: **Scaling Inclusive Mechanization through Rural Entrepreneurship and Financial Innovation**

**Title of Core Flagship:** 2.2 Catalyzing Mechanization and Postharvest Services through Agribusiness Growth

**Secondary Flagships:** 2.1

**Lead Scientist(s) / Contact Person(s) / Center(s):**

**CIMMYT** – Technical and Delivery Lead

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- Dr. Blessing Mhlanga – Cropping Systems Agronomist, CIMMYT
- Dr. Md Abdul Matin – Mechanization Specialist, CIMMYT

**Key CGIAR Center Partners:**

**Alliance of Bioversity International & CIAT** – Access to Finance Lead

- *Pedro Chilambe* – Research Team Leader, Climate Action – Africa Region
- *Dr. Evan Girvetz* – Principal Scientist

**IWMI** – Responsible Scaling and Inclusion Lead

- *Dr. Ojongetakah Enokenwa Baa* (Researcher, Gender, Social Inclusion and Youth)

**Technical Linkages to other Science Programs:**

- Sustainable farming
- Gender Equality and Inclusion
- Innovation Packaging and Scaling Readiness (IPSR)

## Section 1 – Innovation Package, Value Proposition & Alignment

### The problem to be solved

Agriculture in much of Africa is still characterized by labor-intensive practices reliant on manual and animal draft power. In Zambia, recurring droughts and the decimation of draft cattle herds have worsened the problem, leaving smallholder farmers with limited traction sources. This has constrained timely land preparation, crop establishment, harvesting, and postharvest processing. As a result, yields remain low, and the adoption of climate-smart practices and improved inputs to increase productivity is limited. The problem is especially acute for women, youth, and marginalized farmers who lack access to affordable, scale-appropriate mechanization solutions and finance. Mechanization is central not only to reducing drudgery and improving productivity, but also to unlocking climate resilience and inclusivity in food systems. Without appropriate machines, farmers face persistent barriers

to adopting conservation agriculture, irrigation, and postharvest technologies. Existing financial products and services further limit uptake, and unlocking this constraint will enable rapid scaling.

## Your scaling ambition

The ambition of this scaling track is to establish a sustainable and inclusive mechanization ecosystem that reaches scale through entrepreneurship and finance. Building on the Zimbabwe experience, the track proposes a phased approach to establishing 500 additional mechanization service providers by 2026 and, in total, 1000 by 2027 in Zambia, aiming to reach more than 100,000 farmers by 2028 in Southern Africa. The scaling track aims to expand from Zambia to four expansion countries as the project progresses: Zimbabwe, Malawi, Kenya, and Ethiopia, leveraging bilateral funding. Service providers will deliver affordable, climate-smart mechanization services across the value chain: land preparation, planting, transport, and postharvest processing. The model emphasizes youth and women as entrepreneurs, using innovative financial products to de-risk lenders. The ambition is fully aligned with Flagship 2.2 on mechanization as the core flagship, contributing directly to Grand Challenge 2 on delivering inputs and services at scale.

## Innovation & Value Proposition

This Solution Scaling Track expands CIMMYT's **mechanization starter pack** to address the acute lack of affordable, inclusive, climate-smart mechanization for smallholders in East and Southern Africa. The package combines access to low-cost machinery and attachments (e.g., two-wheel tractor, basin digger, ripper, direct seeder, sheller, thresher, trailer, and postharvest/value addition equipment) with **smart lending products**, technical and business training, financial literacy, and entrepreneurial support.

The delivery model is the **Mechanization Service Provider (SP)** approach, where rural entrepreneurs—including youth and women—acquire starter packs through co-financing and deliver fee-based services to farmers. This addresses labor shortages, boosts productivity under climate stress, and creates rural enterprises.

### Value proposition:

- **Farmers:** affordable mechanization reduces drudgery, improves yields, resilience, and postharvest services.
- **Service Providers:** structured entry into rural business ecosystems with reliable income.
- **Financial Institutions:** provide tailored lending products for emerging mechanization entrepreneurs.
- **Institutions:** replicable public–private partnership (PPP) model for scaling mechanization innovations elsewhere

## Customers & Alignment

The main institutional customers are Zambia's Ministry of Agriculture Departments, including Technical and Advisory Services, Agribusiness Management, along with donor-funded programs, NGOs, agriculture commodity buyers, and financial institutions seeking scalable, sustainable mechanization models. End-users are smallholder farmers, cooperatives, and rural micro-entrepreneurs. The model prioritizes women and youth in

selection, training, and financing to ensure inclusion. This solution aligns directly with Zambia’s mechanization strategies and donor priorities on food security and resilience, while offering a practical pathway for scaling. During expansion to other territories, the Ministries of Agriculture in Southern and Eastern Africa and additional scaling partners will be critical.

## Evidence of Impact

Piloted in Zimbabwe under [Ukama Ustawi](#) and [USAID’s Mechanization Activity](#) between 2022 and 2025, the model has established more than 500 Mechanization Service Providers (MSPs) serving 20,000+ farmers with spillover to 35,000+ farmers with strong demand emerging in Zambia and Malawi. These services generated more than US\$40,000 in revenue despite a challenging drought in 2023. Private sector investment exceeded US\$337,000 in Zimbabwe. [Impact stories](#) demonstrate how mechanization empowered women’s groups, such as those operating chopper grinders to produce livestock feed during droughts. Qualitative evidence from [GenderUp and Scaling Readiness workshops](#) and [“Willingness to Pay” studies](#) conducted in the last three years reveals high satisfaction among users and readiness to scale-up mechanization if a machinery supply chain and affordable financial products and services are in place, and awareness is developed. IPSR profiles also confirm that mechanization starter packs are at a high scaling readiness level.

## Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

### Scaling Pathway & Sustainability

Scaling will follow a **blended public–private partnership (PPP) model** designed for expansion and long-term sustainability. The approach combines equity from service providers (SPs), financing from formal institutions, and institutional support to ensure resilience.

- **Financial model:**
  - SPs contribute 30% equity.
  - Financial institutions finance the remaining 70% through loans bundled with climate risk insurance.
  - Repayment schedules align with crop cycles, reducing default risk.
  - A risk-sharing facility (RSF) covers 25–65% of defaults, encouraging lending.
- **Multi-channel financing:**
  - Savings & Credit Cooperative Organizations (SACCOs) and Microfinance Institutions (MFIs) extend affordable loans to SPs, smaller entrepreneurs, and marginalized groups.
  - Hire-purchase schemes with distributors enable gradual repayment and affordability.
  - This blended model, proven under Mechanization and Extension Activity and SDC projects, positions SPs as credible, bankable clients.
- **Operational sustainability:**
  - Regional hubs provide training, spare parts, repairs, and feedback.
  - Rural mechanics training ensures equipment uptime.
  - Ministry of Agriculture officers capacitated on mechanization.

- Embedding services in market systems enables SPs to earn year-round from land preparation, transport, processing, and postharvest services.

#### Scaling milestones:

| Milestone                                                                                 | Target Year | Indicator                                          |
|-------------------------------------------------------------------------------------------|-------------|----------------------------------------------------|
| Affordable finance is made available to procure machinery                                 | 2026        | Financial products on offer                        |
| 500 SPs trained and equipped                                                              | 2026        | Verified by training records and asset tags        |
| 1000 SPs operational                                                                      | 2027        | Training records and asset tags; household surveys |
| 100,000 farmers reached with mechanization services<br>Expanded to 3 additional countries | 2028        | Household surveys<br>MoUs and implementation plans |

This blended pathway guarantees affordability for farmers, profitability for service providers, and resilience for food systems.

### Partnerships & Leverage

The Solution Track is supported by strong demand from diverse external partners to ensure that it is not driven solely by CGIAR but through public and private partnerships, maximizing leverage and sustainability.:

- **National Governments**
  - *Zambia Ministry of Agriculture (and MoAs in spillover countries)*
    - Partner in the Soil and Water Accelerator and CIAT-led financial product.
    - Interested in integrating mechanization bundles for climate-resilient maize.
- **Financial Institutions**
  - *Commercial banks and financing institutions in Zambia* (e.g., ZANACO, AgLeaseCo)
    - Piloted lending to SPs using equipment as collateral, bundled with training and distributor engagement.
    - Exploring collaboration with CIAT and IWMI to operationalize the risk-sharing facility and co-develop bundled financial products.
  - *SACCO networks and MFIs* (Vision Fund, Microloan Foundation) identified as viable channels for scaling local asset financing.
- **Equipment Distributors and Private Sector**
  - Machinery suppliers and manufacturers (CAMCO, SARO, Growmore, Tractor Providers) engaged in supply, co-design, and after-sales service.
  - Several exploring distributor financing.
  - Potential for Hello Tractor-like models for digital scheduling, payment, and maintenance.

- Input dealers and postharvest vendors targeted for bundling (e.g., shellers, dryers, packaging).
- **Aggregators and Cooperatives**
  - Contract farming schemes offer entry points for structured repayment, farmer identification, and co-financing.
  - Aggregators in maize and horticulture keen to integrate mechanization to reduce bottlenecks.

#### Leverage:

- [Ukama Ustawi \(UU\)](#) (Malawi, Zambia, Zimbabwe): regional experience scaling bundles through inclusive approaches, digital/financial advisories, and policy integration. MFAT committed catalytic finance through UU.
- [SIFAZ \(Zambia\)](#) and Water and Soil Accelerator: scaling mechanization with delivery partners (CFU, TLC).
- [AICCRA- Zambia](#): piloted mechanization and solar irrigation innovations in Zambia
- Mechanization and Extension Activity (Zimbabwe): lessons on developing service providers and innovative financing with BancABC.
- Private sector pilots: Service Providers already operating mechanization models in Zambia and Zimbabwe.

#### Risks & Bottlenecks

This risk matrix summarizes the key risks, their likelihood and impact, mitigation measures, and alignment with CGIAR Flagships.

| Risk Category | Description                                                                                                       | Likelihood<br>(1=Low,<br>5=High) | Impact<br>(1=Low,<br>5=High) | Mitigation Measures                                                                                                                            |
|---------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Financial     | Financial institutions perceive smallholder mechanization as high risk due to low collateral and volatile incomes | 4                                | 5                            | Risk-sharing facilities, bundled insurance, and aggregation-based repayment models                                                             |
| Policy        | Shifts in national mechanization strategies that may affect scaling and support structures                        | 3                                | 4                            | Embed service provider networks into government programs                                                                                       |
| Operational   | Lack of spare parts and repair services is limiting service provider efficiency                                   | 3                                | 3                            | Establish hubs and train rural mechanics to provide timely repairs and get feedback for adjustments                                            |
| Social        | Exclusion of women and youth from service provision opportunities                                                 | 2                                | 3                            | Apply inclusive scaling approaches such as GenderUp and GALS tools to prioritize women in postharvest services and youth in repair/innovation. |

|               |                                                             |   |   |                                                                                                 |
|---------------|-------------------------------------------------------------|---|---|-------------------------------------------------------------------------------------------------|
|               |                                                             |   |   | Identification and adapting equipment to be gender-responsive and user-friendly                 |
| Environmental | Potential soil degradation and overuse of tillage practices | 3 | 4 | Link mechanization with conservation agriculture practices (minimum tillage, <i>Pfumvudza</i> ) |

## Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

### Readiness & Responsible Scaling

The mechanization starter pack and service provider model are currently assessed at Scaling Readiness levels five to six, meaning they have moved beyond proof of concept and pilot demonstrations to field expansion. The innovation has been validated under multiple donor-funded projects in Zimbabwe and Zambia, including Ukama Ustawi, SIFAZ, SDC-funded R4 projects, and the USAID Mechanization and Extension Activity, confirming profitability and adoption (500+ SPs in Zimbabwe) under diverse conditions. Each SP can serve 50–150 farmers depending on seasonal workload and tasks. Responsible scaling is embedded at several levels:

- **First:** awareness, SP selection, and training explicitly include women and youth, ensuring inclusion and equitable employment.
- **Second:** mechanization services are tied to sustainable practices, linking equipment to conservation agriculture and climate-smart irrigation.
- **Third:** digital booking and repayment systems enhance accountability, transparency, and monitoring.
- **Finally:** SP networks, financiers, suppliers, and government partners create feedback loops for continuous learning and adaptation.

### Achievable Target, Resources & Activities

#### Targets

- Establish 500 Mechanization Service Providers (MSPs) in Zambia by 2026.
- Expand to 1,000 MSPs in Zambia by 2027.
- Spillover into Zimbabwe, Malawi, Kenya, and Ethiopia by 2028, leveraging bilateral funding to reach 100,000 farmers with mechanization services.

#### Resources

We anticipate USD 500,000/year (2026–2028) from S4I, strategically complemented by bilateral/W3 investments (e.g., SIFAZ, AICCRA, Water & Soil Accelerator). Distribution:

- **CIMMYT (USD 300,000):** Awareness, SP recruitment, hub development, training.
- **ABC (USD 120,000):** Develop financial products, deploy digital tools, climate risk support, digital advisory.
- **IWMI (USD 80,000):** Inclusive scaling, gender action plans, gender-lens investment support.

- **Partners (USD 100,000 leveraged):** Bilateral, national public/private sector engagement.

### Activities

- **Recruitment/training (20%):** MSP skills in equipment, maintenance, and business; inclusive scaling training with public-private partners using tailored tools; design gender accelerator action plans with technical support to agripreneurs and SPs in mechanization value chains; and promote gender-lens investments for SPs and agripreneurs.
- **Service hubs (20%):** Training, demos, repairs, spare parts, link private sector with mechanization activities.
- **Finance (30%):** Loans via banks, SACCOs, MFIs with RSFs/insurance; financial literacy; hire-purchase schemes.
- **Awareness (20%):** Field demos and campaigns.
- **Policy (10%):** Embed mechanization into subsidies and extension.

### Outcome

- Mechanization Service Provider models expanded and mainstreamed in Zambia (and neighboring countries)
- Expanded access and increased utilization of mechanization services for smallholder farmers
- Increased capacity amongst Service Providers and farmers
- Increased household productivity, incomes, and strengthened resilience of smallholder farmers through access to mechanization services
- Financial services developed and expanded in Zambia and regionally, transforming smallholder agriculture and building sustainable mechanization ecosystems.

### Timeline, Milestones & Monitoring

The scaling track follows a sequenced pathway to ensure systematic growth, regional replication, and robust monitoring. Annual milestones will be clearly defined, documented, and verified through multiple data sources, including training records, financing agreements, GIS-based service tracking, and household-level surveys.

- **2026:** Establish and equip 500 mechanization service providers in Zambia. Provide inclusive training and gender action plans. Verify progress through training records, machine asset tagging, and financing agreements with partner banks. Service hubs and mechanic training centers will also be operational by this stage, ensuring sustainability and local repair capacity.
- **2027:** Recruit additional MSPs; scale mechanization services to smallholder farmers, delivering bundled solutions in land preparation, crop establishment/management, and postharvest; explore use of GIS-based service logs to track geographical reach; household surveys to assess adoption and satisfaction, and repayment records from financial institutions to confirm business viability.
- **2028:** Expand the model into **Zimbabwe, Malawi, Kenya, and Ethiopia**, supported by formal MoUs, national partnerships, and implementation plans. At this stage,

second-generation financial products and irrigation packages will also be piloted to increase inclusivity and climate resilience.

Monitoring and learning systems will emphasize adaptive management, ensuring that lessons from earlier years inform adjustments and strengthen regional replication.

## Optional Annexes

1. IPSR Innovation Profile – [Mechanization Starter Pack](#)
2. Scaling Readiness results ([Workshops, 2023–2024](#))
3. GenderUp participatory scaling design ([IWMI, 2023](#))
4. Mechanization Activity Final Report ([June 2025](#))
5. Mechanization Impact Stories (2025) <https://shorturl.fm/NVwQV>;  
<https://hdl.handle.net/10568/162638>;
6. Publications: [Ngoma et al. 2023. Willingness to Pay](#); <https://tinyurl.com/mrypv478>
7. Letters of Intent – BancABC, CBZ, Ministries of Agriculture - available on request
8. Mechanization finance business case: <https://hdl.handle.net/10568/163770>
9. Supply chain partner engagement: <https://hdl.handle.net/10568/168223>
10. Detailed investment case: <https://hdl.handle.net/10568/168232>